

PD Group:	Name:
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**ANDERSON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS**

**BIOLOGY
Higher 2**

9648/03

16 SEPT 2011 FRIDAY

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

A **NIL** return is required.

Write your answers in the spaces provided on the question paper.

All working for numerical answers must be shown.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This document consists of **13** printed pages.

Section A

Answer **all** the questions in this section.

- 1 (a) Compare the differences on how a genomic library and a cDNA library are produced.

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[2]

- (b) Cancer cells express proteins called markers on their surface. Each marker is unique to each type of cancer cells, and hence markers are useful in distinguishing different cancer cell types. A cancer marker has been discovered in a patient, and its cDNA level is traced over the course of 12 weeks as shown in Fig. 1.1.

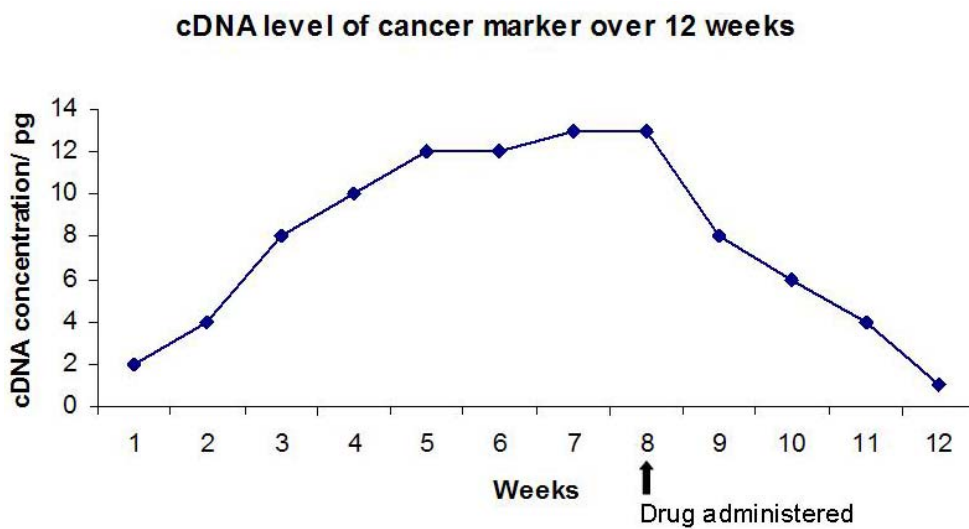


Fig. 1.1

- (i) Explain the advantages of using a cDNA library over a genomic DNA library in studying the expression level of this cancer marker.

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(ii) Describe and explain the change in cDNA in the first eight weeks.

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 [2]

(iii) Suggest how data such as those given in Fig. 1.1 is useful to cancer scientists.

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 [1]

(c) In order to study the cancer marker protein structure, a scientist wants to clone the cDNA into a plasmid as shown in Fig. 1.2.

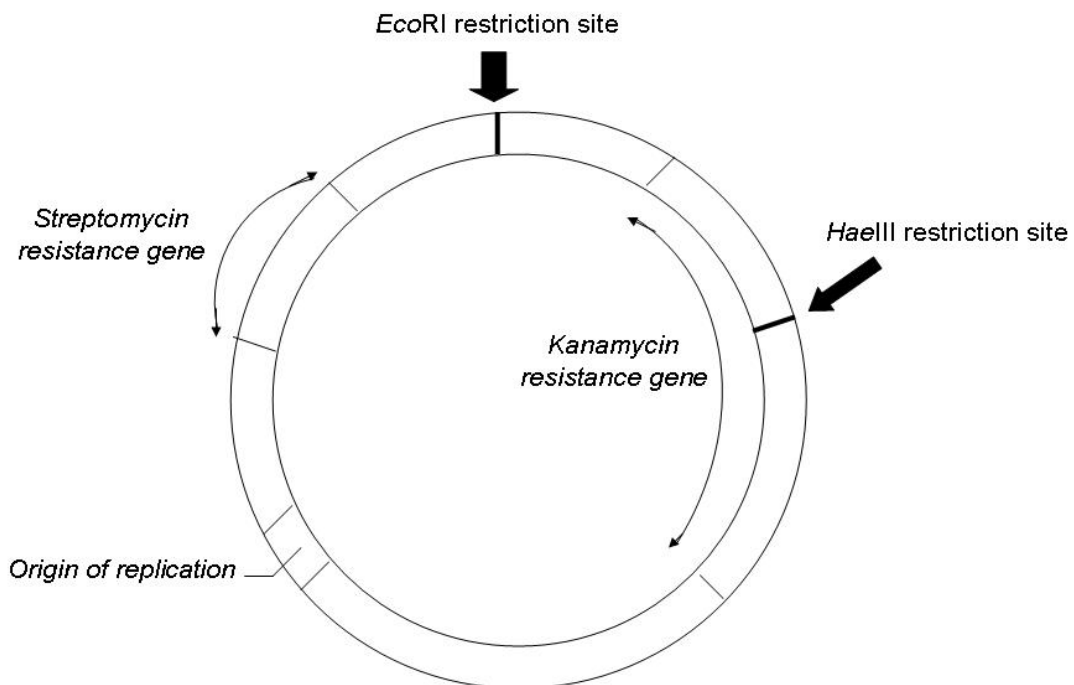


Fig. 1.2

The scientist has the following items:

- Plasmid vector shown in Fig. 1.2,
- cDNA of the cancer marker gene (with blunt ends),
- Four restriction enzymes as shown in Fig. 1.3,

Restriction enzymes	Restrictions sites (arrows represent cut made by enzyme)
(i) <i>AluI</i>	$ \begin{array}{c} \downarrow \\ 5' \dots A G C T \dots 3' \\ 3' \dots T C G A \dots 5' \\ \uparrow \end{array} $
(ii) <i>HaeIII</i>	$ \begin{array}{c} \downarrow \\ 5' \dots G G C C \dots 3' \\ 3' \dots C C G G \dots 5' \\ \uparrow \end{array} $
(iii) <i>BamHI</i>	$ \begin{array}{c} \downarrow \\ 5' \dots G G A T C C \dots 3' \\ 3' \dots C C T A G G \dots 5' \\ \uparrow \end{array} $
(iv) <i>EcoRI</i>	$ \begin{array}{c} \downarrow \\ 5' \dots G A A T T C \dots 3' \\ 3' \dots C T T A A G \dots 5' \\ \uparrow \end{array} $

Fig. 1.3

- DNA ligase,
- four types of linker DNA molecules with restrictions sites as shown in Fig. 1.3.

Explain how the scientist could clone the cDNA into the plasmid which would eventually lead to the selection of the bacteria containing the recombinant plasmid.

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- (d) Even with the recombinant plasmid transformed into the bacteria, the gene of interest from the cDNA library may not be expressed correctly. Suggest two reasons why.

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[Total: 15]

- 2 Only the coding DNA strand of a gene is normally transcribed to mRNA. The complementary non-coding DNA strand is not normally transcribed. By inserting a promoter sequence at the end of a non-coding strand, thus forming an antisense gene, transcription can occur. The product anti-mRNA binds the normal mRNA to form double-stranded RNA (duplex RNA), which cannot be translated by ribosomes. This technique can be used to suppress specific genes. The sequence of events is shown in Fig. 2.1.

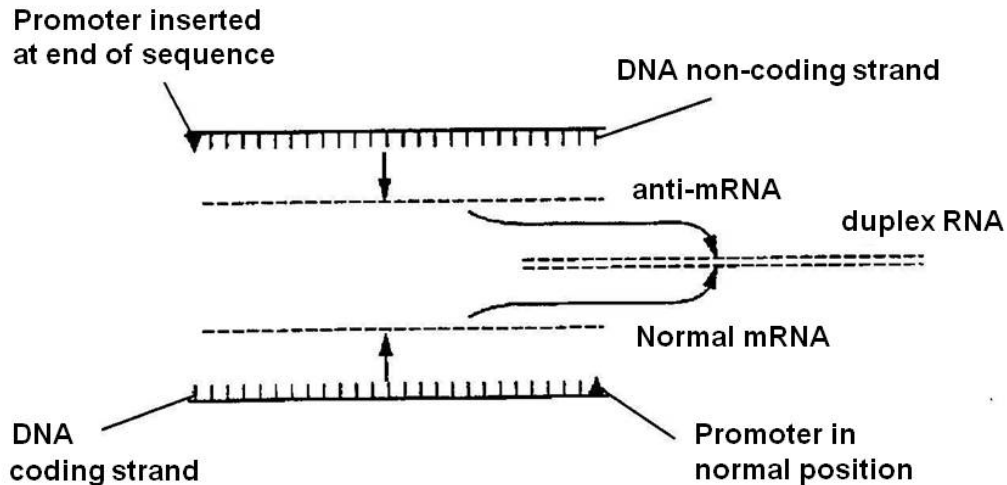


Fig. 2.1

- (a) Explain why the resulting duplex RNA cannot be translated.

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[2]

- (b) The insertion of a promoter into the antisense strand of DNA involves the formation of recombinant DNA. Once the recombinant DNA is formed, it is inserted into plant cells obtained from a protoplast culture.

Protoplasts are fragile and easily damaged. The culture medium is maintained at very low water potential in order not to destroy the protoplasts.

Suggest why a low water potential must be maintained in the culture medium when culturing protoplasts.

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[2]

- (c) A genetically modified tomato, known as Flavr Savr, is the first genetically modified food to be commercially sold for human consumption. This tomato is made more rot-resistant by adding an antisense gene to interfere with the production of the enzyme polygalacturonase (PG). The enzyme causes the softening of the fruit by degrading pectin and this results in the fruit being more susceptible to being damaged by fungal infection

PG production during fruit ripening was measured in plants with one and with two antisense genes, and in normal plants. The results are shown in Fig. 2.2.

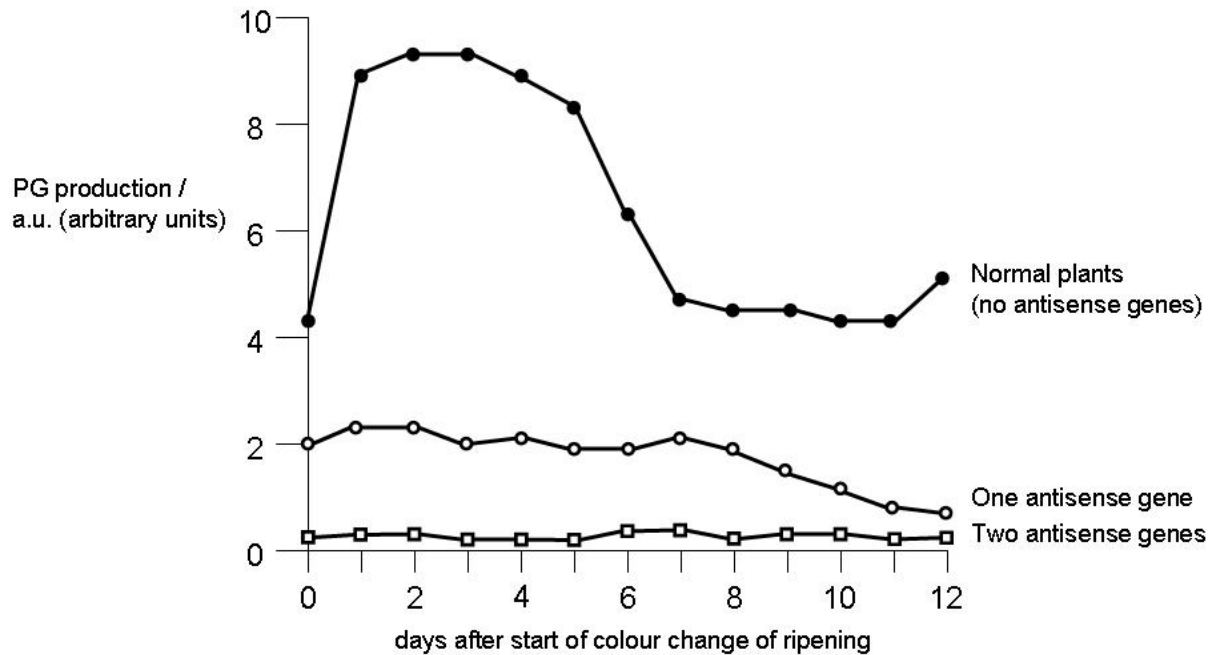


Fig. 2.2

With reference to Fig. 2.2, describe and explain the difference in PG production between

- (i) normal tomato plants and transformed plants with one antisense gene.

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- (ii) transformed plants with one antisense gene and those with two antisense genes.

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- (d) Genetic engineering can also be used to introduce genes from other species into crop plants. This has led to the production of crops such as Bt corn and golden rice.

Between the years 1900 and 2000, crop yields across the world increased by almost 300%.

Discuss why such genetic engineering has contributed little to this rise in yield and explain your answer.

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[3]

[Total: 12]

- 3 Cystic fibrosis (CF) is one of the most common lethal inherited diseases in humans. There are approximately 50,000 sufferers worldwide, with an additional 10 million carriers. CF is a result of a single gene mutation and demonstrates autosomal recessive inheritance.

Fig. 3.1 shows the inheritance of the disease in a family in which both parents are carriers.

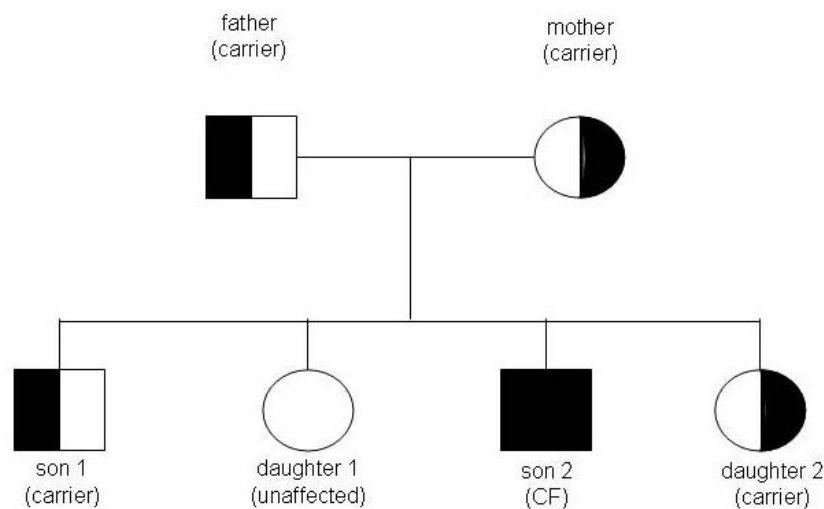


Fig. 3.1

- (a) Explain how the diagram in Fig. 3.1 shows that the disease is autosomal recessive.

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- (b) The maps of the sites for restriction enzyme R in the wild type and the mutated cystic fibrosis genes are shown schematically in Fig. 3.2.

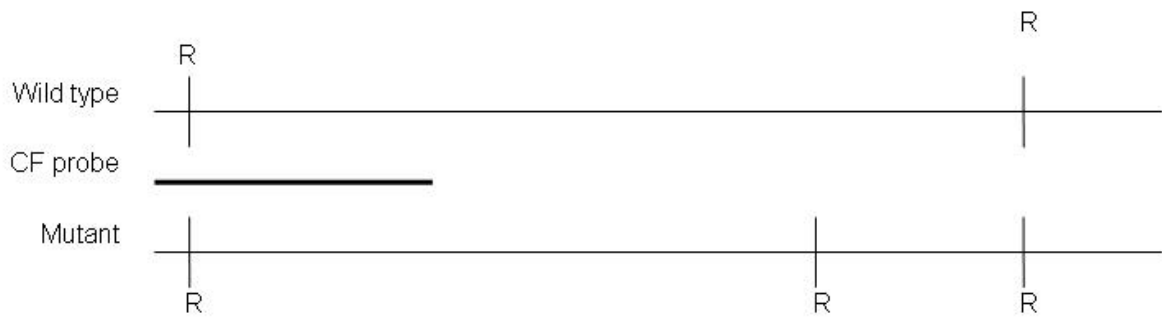


Fig. 3.2

Samples of DNA obtained from a fetus (F) and her parents (M and P) were analyzed by gel electrophoresis followed by the Southern blot technique and hybridization with the radioactively labelled probe designated "CF probe" in Fig. 3.2. The autoradiograph results are shown in Fig. 3.3.

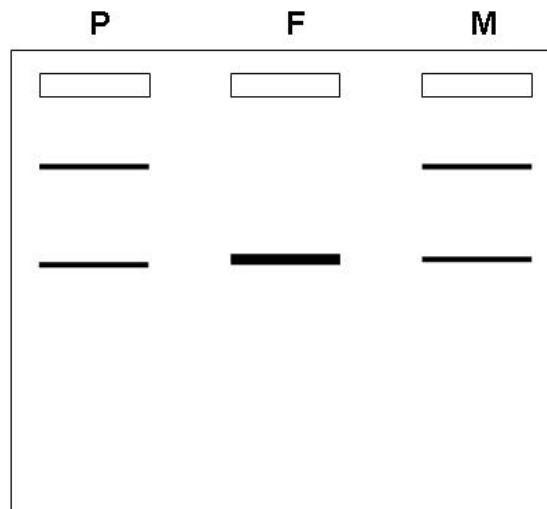


Fig. 3.3

With reference to Fig. 3.3, explain the banding pattern of the fetus.

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- (c) The CF gene is found on chromosome 7 and in non-sufferers it gives rise to a protein called cystic fibrosis transmembrane conductance regulator (CFTR). The defect which most commonly leads to the disease involves the deletion of three nucleotides from the gene. This is illustrated in Fig. 3.4.

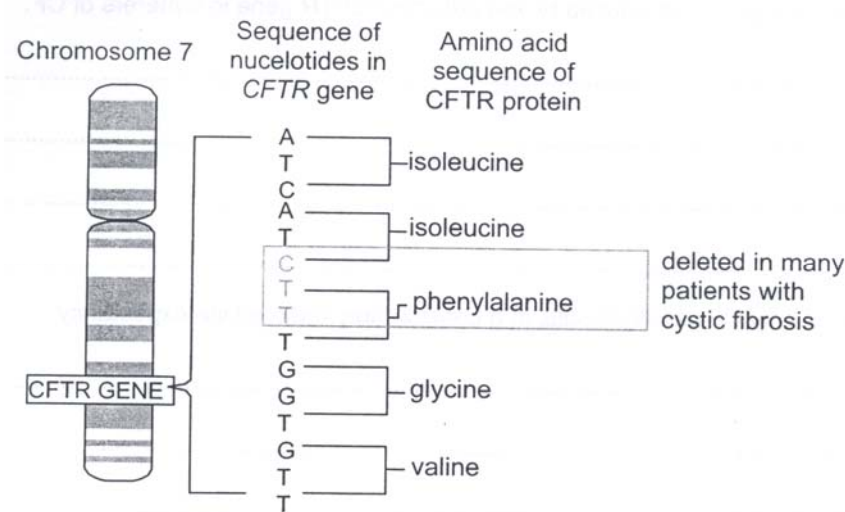


Fig. 3.4

- (i) With reference to Fig. 3.4, explain how this deletion results in the production of a defective protein.

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- (ii) The protein produced by the CFTR gene is involved in the transport of chloride ions across cell membranes. In sufferers of CF there is a build up of sodium and chloride ions that causes the body to produce a thick, sticky mucus. Most individuals with the disease have a considerably reduced life expectancy.

Outline the problems caused by the defective CFTR gene in sufferers of CF.

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[Total: 13]

A colorimeter is a device that can be used to measure the concentration of a solute by measuring the absorbance of different wavelengths of light in a solution. Generally, a solution of higher concentration will absorb more light energy at a specific wavelength. Therefore, the higher the absorbance value, the higher the concentration of a solution. Although absorbance has no units, it is often recorded as absorbance units (A.U.)

[Total: 12]

- a solution containing an unknown concentration of sucrose extracted from the new variety of vegetable, labelled **P**.
- a 10% sucrose solution, labelled **S**, which can be diluted to give various solutions of known sucrose concentration.

- a description of the method used including the scientific reasoning behind the method,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

[Total: 20]